

Pokhara University

Level: Bachelor

Semester: xxxx

Year: xxxx

Program: BE

Full Marks: 100

Course: Calculus I

Time: 3 hrs.

Pass Marks: 45

MODEL QUESTION

Candidates are required to give their answer in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all questions.

1. a) If $y^{1/m} + y^{-1/m} = 2x$ then prove that: 8
$$(x^2 - 1)y_{n+2} + (2n + 1)xy_{n+1} + (n^2 - m^2)y_n = 0.$$

b) If $f(x) = (x - a)^m(x - b)^n$; where, m and n are positive integer, 7
show that c in Rolle's theorem divides the segment $a \leq x \leq b$ in the ratio $m : n$.

OR

Assuming the validity of expansion, find Maclaurin's series expansion of $f(x) = \log(\sec x)$ and use it for expansion of $\tan x$.

2. a) Find all the asymptotes of the curve $x^2(x - y)^2 - a^2(x^2 + y^2) = 0$. 8
b) Find the radius of the curvature $x = a(t + \sin t)$, $y = a(1 - \cos t)$ at any point. 7
3. Attempt any **three** of the following: [**3 × 5 = 15**]
a) Evaluate: $\int \frac{1}{5 \sin x + 4} dx$.
b) Show that: $\int_0^{\pi/4} \log(1 + \tan \theta) d\theta = \frac{\pi}{8} \log 2$
c) Evaluate: $\int_1^{\infty} \frac{\log x}{x^2} dx$.
d) Evaluate by using gamma function $\int_0^{\pi/6} \cos^4 3\theta \sin^2 6\theta d\theta$.
4. a) Find the volume of the solid generated by revolving the region bounded 8
by the parabola $y = x^2 + 1$ and the line $y = x + 3$ about x-axis.

- b) State Euler's theorem for homogeneous function of three variables with degree n . If $u = \frac{y}{x} + \frac{z}{x} + \frac{x}{y}$, show that:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0.$$

7

5. a) Find the stationary point of the function $f(x,y,z) = x^2 + xy + y^2 + 3z^2$ under the constraint $x + 2y + 4z = 60$. Also find the extreme value. 7
- b) Solve: $x \frac{dy}{dx} + y = y^2 \log x$. 8

OR

A tank initially contains 4 kg of salt dissolved into 100 litres of water. A solution of 2 kg of salt per litre is allowed to enter the tank at the rate of 5 litres per minute and uniform solution is drained from the tank at the same rate. Find the amount of salt at any time. Also determine the salt in tank in 10 minutes.

6. a) Find the solution of initial value problem: 7

$$x^2 y'' - 4xy' + 6y = 0; y(1) = \frac{3}{2}, y'(1) = 1.$$

8

- b) Solve the differential equation: $y'' + 5y' = 15x^2$.

7. Attempt all the questions: **[4 × 2.5 = 10]**

- a) Show that function $f(x) = \begin{cases} x^2 + 3 & \text{for } x < 1 \\ 3x + 1 & \text{for } x \geq 1 \end{cases}$

is not differentiable at $x = 1$.

- b) Find the arc length of the curve $y = x^{\frac{3}{2}}$ from $x = 0$ to $x = 2$.

- c) Solve: $\sqrt{1-x^2} dy + \sqrt{1-y^2} dx = 0$

- d) If $z = x^3 - y^3$, $x = \frac{1}{t+1}$, and $y = \frac{t}{t+1}$, find $\frac{dz}{dt}$.

